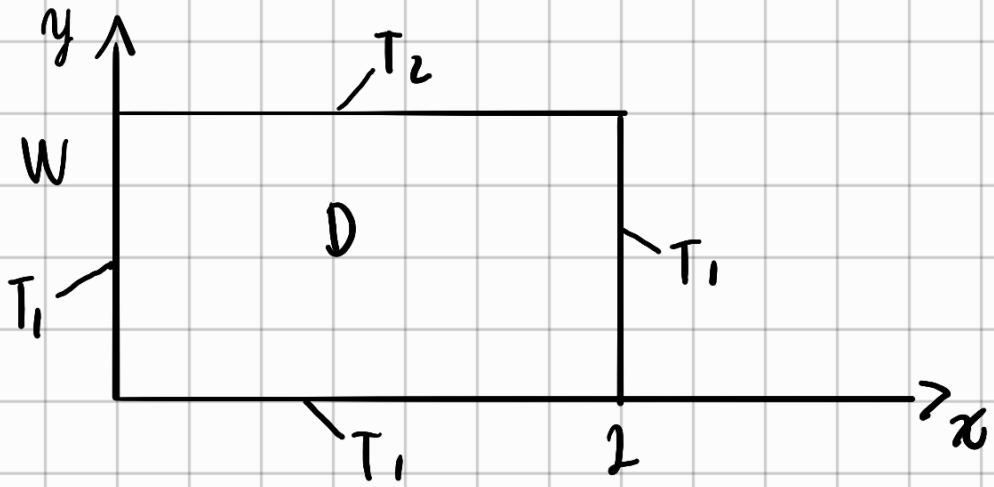
Expo 3:

- Caso analítico resoluble en 2D (estado estacionario)



$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Con las condiciones de contorno

$$\begin{cases} T(0 \le x \le L, y = 0) = T(x = 0, 0 \le y \le w) \\ = T(x = L, 0 \le y \le W) = T_1 \end{cases}$$

$$\{ T(0 \le x \le L, y = w) = T_2$$

Sea $\Theta = \dfrac{T - T_1}{T_2 - T_1}$ esta función se toma para que cuando $T = T_1 \rightsquigarrow \Theta = 0$ y cuando $T = T_2 \rightsquigarrow \Theta = 1$

Ahora como $\Theta(x,y)$ entonces $\Theta(x,y) = X(x)Y(y)$

$$\nabla^2 T = 0 \rightsquigarrow \nabla^2 \Theta = 0 \rightsquigarrow \frac{\partial^2 (XY)}{\partial x^2} + \frac{\partial^2 (XY)}{\partial y^2} = 0$$

$$\rightsquigarrow Y \frac{\partial^2 X}{\partial x^2} + X \frac{\partial^2 Y}{\partial y^2} = 0$$

$$\rightsquigarrow \frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = 0$$

$$\rightarrow \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = Cte = \lambda^2$$

Aparecen 2 ecuaciones

- $$\frac{d^2 X}{d x^2} + X \lambda^2 = 0$$

$$\rightsquigarrow \text{Sea } X = e^{rx} \rightsquigarrow \frac{d^2 X}{d x^2} = r^2 e^{rx}$$

$$\rightsquigarrow r^2 e^{rx} + e^{rx} \lambda^2 = 0 \rightsquigarrow e^{rx}(r^2 + \lambda^2) = 0$$

$$r^2 + \lambda^2 = 0 \rightsquigarrow r = \sqrt{-\lambda^2} = \pm i\lambda$$

$$\rightsquigarrow x = C_1 e^{i\lambda x} + C_2 e^{-i\lambda x}$$

$$\rightsquigarrow x = C_1 (\cos(\lambda x) + i \operatorname{sen}(\lambda x)) + C_2 (\cos(\lambda x) - i \operatorname{sen}(\lambda x))$$

$\rightarrow X = C_1 \cos(\lambda x) + C_1 i \operatorname{sen}(\lambda x) + C_2 \cos(\lambda x)$
$- C_2 i \operatorname{sen}(\lambda x)$

$\rightsquigarrow X = \underbrace{(C_1 + C_2)}_{A} \cos(\lambda x) + \underbrace{(C_1 i - C_2 i)}_{B} \operatorname{sen}(\lambda x)$

$\rightarrow X(x) = A \cos(\lambda x) + B \operatorname{sen}(\lambda x)$

$\circ \dfrac{d^2 Y}{dy^2} - \lambda^2 Y = 0$

$\rightsquigarrow$ Sea $Y = e^{\nu y} \rightsquigarrow \dfrac{d^2 Y}{dy^2} = \nu^2 e^{\nu y}$

$\rightsquigarrow \nu^2 e^{\nu y} - \lambda^2 e^{\nu y} = 0 \rightarrow e^{\nu y}(\nu^2 - \lambda^2) = 0$

$\rightsquigarrow \nu^2 - \lambda^2 = 0 \rightsquigarrow \nu^2 = \lambda^2 \rightsquigarrow \nu = \pm \lambda$

$\rightsquigarrow Y = C_1 e^{\lambda y} + C_2 e^{-\lambda y}$

$$\begin{cases} \operatorname{senh}(z) = \dfrac{e^z - e^{-z}}{2} \rightsquigarrow e^z = \cosh(z) + \operatorname{senh}(z) \\ \cosh(z) = \dfrac{e^z + e^{-z}}{2} \rightarrow e^{-z} = \cosh(z) - \operatorname{senh}(z) \end{cases}$$

$\rightarrow Y = C_1 [\cosh(z) + \operatorname{senh}(z)] + C_2 [\cosh(z) - \operatorname{senh}(z)]$

$\rightarrow Y = C_1 \cosh(z) + C_1 \operatorname{senh}(z) + C_2 \cosh(z) - C_2 \operatorname{senh}(z)$

$\rightarrow Y = C_1 \cosh(z) + C_1 \operatorname{senh}(z) + C_2 \cosh(z) - C_2 \operatorname{senh}(z)$

$\rightarrow Y = \underbrace{(C_1 + C_2)}_{C} \cosh(z) + \underbrace{(C_1 - C_2)}_{D} \operatorname{senh}(z)$

$\rightarrow Y = C \cosh(\lambda y) + D \operatorname{senh}(\lambda y)$

Ahora aplicamos Condiciones de Contorno

[Sea $\Theta = \frac{T - T_1}{T_2 - T_1}$ esta función se toma para que Cuando $T = T_1 \rightsquigarrow \Theta = 0$ y Cuando $T = T_2 \rightsquigarrow \Theta = 1$]

- $\Theta(0, y) = 0 \rightsquigarrow \Theta(0, y) = X(0)\, Y(y) = 0$
$\rightsquigarrow X(0) = 0 \rightarrow X(0) = A \cos(0) + B \operatorname{sen}(0) = A$
$\rightarrow A = 0$

- $\Theta(L, y) = 0 \rightsquigarrow X(L) = 0 \rightarrow X(L) = B \operatorname{sen}(\lambda L) = 0$
$\rightarrow \operatorname{sen}(\lambda L) = 0$

Como $\operatorname{sen}(n\pi) = 0 \rightsquigarrow \lambda_n = \frac{n\pi}{L}$

$$\rightarrow \Theta(x, y) = \sum_{n=1}^{\infty} B_n \operatorname{sen}\left(\frac{n\pi x}{L}\right) Y_n(y)$$

- $\Theta(x,0)=0$

$\rightarrow Y(y) = C\,\text{senh}(\lambda y) + D\cosh(\lambda y)$

$\rightarrow Y(0) = C\,\text{senh}(0) + D\cosh(0) = D = 0$

$\rightarrow Y(y) = C\,\text{senh}(\lambda y)$

- $\Theta(x,W) = 1$

$\rightarrow \Theta(x,W) = \sum_{n=1}^{\infty} B_n \,\text{sen}\left(\frac{n\pi x}{L}\right) C_n \,\text{senh}\left(\frac{n\pi W}{L}\right)$

notemos que $\forall x$ $\Theta(x,W)=1$ entonces
$A_n = B_n C_n \,\text{senh}\left(\frac{n\pi W}{L}\right)$

$\rightsquigarrow 1 = \sum_{n=1}^{\infty} A_n \,\text{sen}\left(\frac{n\pi x}{L}\right)$ notemos que A_n se comporta como un coeficiente de fourier

$A_n = \frac{2}{L}\int_0^L 1 \,\text{sen}\left(\frac{n\pi x}{L}\right)dx \rightarrow A_n = \frac{2}{L}\left[-\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right]_0^L$

$\rightarrow A_n = \frac{2}{L}\left[-\frac{L}{n\pi}\cos(n\pi) + \frac{L}{n\pi}\right]$

$\rightarrow A_n = \frac{2}{L}\left[\frac{-L}{n\pi}\left[\cos(n\pi) - 1\right]\right] = \frac{-2}{n\pi}\left[(-1)^n - 1\right]$

$= \frac{2}{n\pi}\left[-(-1)^n + 1\right] = \frac{2}{n\pi}\left[(-1)^{n+1} + 1\right]$

$$A_n = D_n \operatorname{senh}\left(\frac{n\pi W}{L}\right) \quad \text{Con} \quad D_n = B_n C_n$$

$$D_n = \frac{A_n}{\operatorname{senh}\left(\frac{n\pi W}{L}\right)}$$

Fórmula final

$$\Theta(x,y) = \sum_{n=1}^{\infty} D_n \operatorname{sen}\left(\frac{n\pi x}{L}\right) \operatorname{senh}\left(\frac{n\pi y}{L}\right)$$

$$= \sum_{n=1}^{\infty} A_n \operatorname{sen}\left(\frac{n\pi x}{L}\right) \frac{\operatorname{senh}\left(\frac{n\pi y}{L}\right)}{\operatorname{senh}\left(\frac{n\pi W}{L}\right)}$$

$$\rightsquigarrow \Theta(x,y) = \frac{2}{\pi} \sum_{n}^{\infty} \frac{(-1)^{n+1}+1}{n} \operatorname{sen}\left(\frac{n\pi x}{L}\right) \frac{\operatorname{senh}\left(\frac{n\pi y}{L}\right)}{\operatorname{senh}\left(\frac{n\pi W}{L}\right)}$$